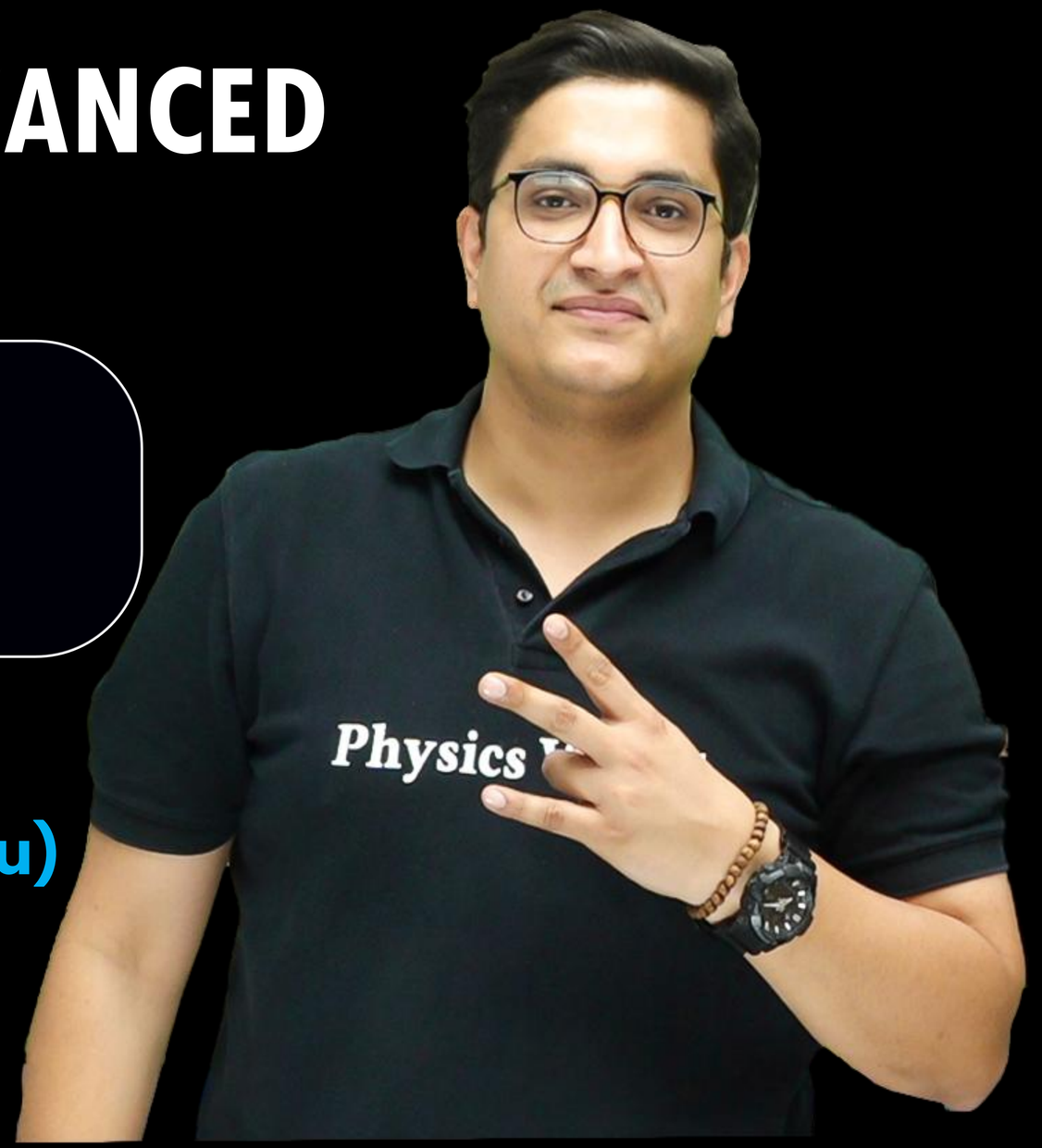
Arjuna JEE 2027

JEE MAINS & IIT JEE ADVANCED

MATHEMATICS

SET THEORY



DIBY (DO IT BY YOURSELF)

By: Sachin Jakhar (Defender Mamu)

DIBY-01

State whether the following collections is a set or not?

(i) The collection of natural numbers between 2 and 20

(ii) The collection of numbers which satisfy the equation $x^2 - 5x + 6 = 0$

(iii) The collection of prime numbers between 1 and 100.

(iv) The collection of all intelligent women in Jalandhar

[Ans: (i) Yes, (ii) Yes, (iii) Yes, (iv) No]

DIBY-02

Write the following set in builder form

(i) Set of all rational number

(ii) $\{2, 5, 10, 17, 26, 37, \ldots \ldots \ldots$

[Ans: (i) $\left\{x : x = \frac{p}{q}, p \in I, q \in N\right\}$

(ii) $\{x : x = \lambda^2 + 1, \lambda \in N\}$]

DIBY-03

Consider set $A = \{1, 2, 3, 4, 5, 6, 7\}$ then:

(i) number of subsets of set A

(ii) number of proper subsets of set A

(iii) number of non-empty subsets of set A.

(iv) number of non-empty proper subsets of set A.

(v) number of non-empty subsets containing only odd numbers.

(vi) number of subsets of A whose product of elements is even.

(vii) number of subsets of A whose product of elements is odd.

[Ans: (i) 128, (ii) 127, (iii) 127, (iv) 126, (v) 15, (vi) 112, (vii) 15]

DIBY-04

Two finite sets have m and n elements respectively. The total number of subsets of first set is 992 more than the total number of subsets of the second set. Find the value of m & n.

[Ans: $m = 10, n = 5$]

DIBY-05

Let $A = \{3, 6, 9, 12, 15, 18, 21\}$, $B = \{4, 8, 12, 16, 20\}$, $C = \{2, 4, 6, 8, 10, 12, 14, 16\}$ and $D = \{5, 10, 15, 20\}$

I $(A \cup B) - C = \{3, 9, 18, 21\}$

II $(A - B) \cap (C - D) = \{6\}$

III $(A \cap B \cap C) \cup D = \{12, 5, 15, 10, 20\}$

Which of the following is correct?

(A) Only I and II

(B) Only II and III

(C) Only III and I

(D) None of these

[Ans. : (B)]

DIBY-06

If A and B are two sets such that $n(A) = 70, n(B) = 60$, and $n(A \cup B) = 110$, then $n(A \cap B)$ is equal to

(A) 240

(B) 50

(C) 40

(D) 20

[Ans. : (D)]

DIBY-07

Let $n(U) = 700, n(A) = 200, n(B) = 300$ and $n(A \cap B) = 100$ then $n(A^c \cap B^c)$

(A) 400

(B) 600

(C) 300

(D) 200

[Ans. : (C)]

DIBY-08

A market research group conducted a survey of 1000 consumers and reported that 720 consumers liked product A and 450 consumers liked product B. What is the least number that must have liked both products?

(A) 170 **(B)** 280 **(C)** 220 **(D)** None of these

[Ans. : (A)]

DIBY-09

If A, B and C are any three sets, then $A - (B \cup C)$ is equal to

(A) $(A - B) \cup (A - C)$ **(B)** $(A - B) \cap (A - C)$

(C) $(A - B) \cup C$ **(D)** $(A - B) \cap C$

[Ans. : (B)]

DIBY-10

If A, B and C are non-empty sets, then $(A - B) \cup (B - A)$ equals

(A) $(A \cup B) - B$ **(B)** $A - (A \cap B)$ **(C)** $(A \cup B) - (A \cap B)$ **(D)** $(A \cap B) \cup (A \cup B)$

[Ans. : (C)]

DIBY-11

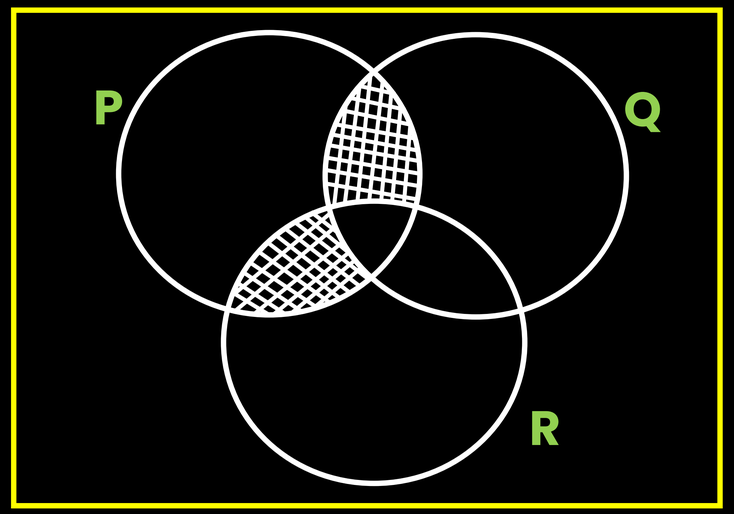
What does the shaded portion of the Venn diagram given below represent?

(A) $(P \cap Q) \cap (P \cap R)$

(B) $\big((P \cap Q) - R\big) \cup \big((P \cap R) - Q\big)$

(C) $((P \cup Q) - R) \cap ((P \cap R) - Q)$

(D) $((P \cap Q) \cup R) \cap ((P \cup R) - R)$



[Ans. : (B)]

DIBY-12

In a class of 55 students, the number of students studying different subjects are 23 in Mathematics, 24 in Physics, 19 in Chemistry, 12 in Mathematics and Physics, 9 in Mathematics and Chemistry, 7 in Physics and Chemistry and 4 in all the three subjects. The number of students who have taken exactly one subject is

(A) 6 **(B)** 9 **(C)** 7 **(D)** 22

[Ans. : (D)]

DIBY-13

If $X = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}, A = \{x \in N: 30 < x^2 < 70\}$, $B = \{x : x$ is a prime number less than 10$\}$, then which of the following is incorrect?

(A) $A \cup B = \{2, 3, 5, 6, 7, 8\}$ **(B)** $A \cap B = \{7, 8\}$

(C) $A - B = \{6, 8\}$ **(D)** $A \Delta B = \{2, 3, 5, 6, 8\}$

[Ans. : (B)]

DIBY-14

If A and B be two sets and $A \cup B \cup C = U$. Then, $((A - B) \cup (B - C) \cup (C - A))'$ is equal to

(A) $A \cup B \cup C$ **(B)** $A \cup (B \cap C)$ **(C)** $A \cap B \cap C$ **(D)** $A \cap (B \cup C)$

[Ans. : (D)]

DIBY-15

If there are three athletic teams in a school 21 are in the basketball team, 26 in hockey team and 29 in the football team. 14 play hockey and basketball, 15 play hockey and football, 12 play football and basketball and 8 play all the games. Then total number of members are

(A) 42 **(B)** 43 **(C)** 45 **(D)** None of these

[Ans. : (B)]

DIBY-16

Let A, B, and C be three sets such that $n(A) = 40, n(B) = 20$, $n(C) = 30, n(A \cap B) = 8, n(B \cap C) = 10, n(C \cap A) = 12$, and $n(A \cap B \cap C) = 5$. Find the number of elements in exactly one of the three sets.

(A) 35 (B) 40 (C) 45 (D) 50

[Ans. : (C)]

DIBY-17

Consider the sets $A = \{n \in \mathbb{N} \mid n$ is a factor of 36$\}$ and $B = \{n \in \mathbb{N} \mid n$ is a factor of 48$\}$. Find $n(A \Delta B)$.

(A) 4 (B) 6 (C) 7 (D) 9

[Ans. : (C)]

DIBY-18

In a class of 50 students, 30 like coffee, 25 like tea, and 15 like both. How many students like neither coffee nor tea?

(A) 5 (B) 10 (C) 15 (D) 20

[Ans. : (B)]

DIBY-19

If $\Pi(A)$ denotes the power set of A, and $A = \{1, 2, \{3, 4\}\}$, find $n(\Pi(A))$.

(A) 4 (B) 8 (C) 16 (D) 32

[Ans. : (B)]

DIBY-20

Let $X = \{1, 2, 3, \ldots, 10\}$ and $A = \{1, 2, 3, 4, 5\}$. How many subsets B of X exist such that $B \cap A = \emptyset$?

(A) 16 (B) 32 (C) 64 (D) 128

[Ans. : (B)]

DIBY-21

Given $n(U) = 100, n(A) = 40, n(B) = 50, n(A \cap B) = 15$.
Match the quantities in Column-I with their values in Column-II.

(A) $A \to p, B \to q, C \to r, D \to s$
(B) $A \to p, B \to r, C \to q, D \to s$
(C) $A \to s, B \to r, C \to p, D \to q$
(D) $A \to s, B \to p, C \to r, D \to q$

Column-I		Column-II	
A.	$n(A \cup B)$	p.	25
B.	$n(A')$	q.	35
C.	$n(A \setminus B)$	r.	60
D.	$n(B \setminus A)$	s.	75

[Ans. : (C)]

DIBY-22

Let A, B, and C be sets such that $n(A) = 10, n(B) = 15$, $n(C) = 20, n(A \cap B) = 4, n(B \cap C) = 6, n(A \cap C) = 5$, and $n(A \cup B \cup C) = 32$. Find $n(A \cap B \cap C)$.

[Ans. : 2]

DIBY-23

Given the sets $A = \{1, 2, 3\}, B = \{3, 4\}, C = \{4, 5, 6\}$, then $A \cup (B \cap C)$ is

(A) {3} (B) {1, 2, 3, 4} (C) {1, 2, 4, 5} (D) {1, 2, 3, 4, 5, 6}

[Ans. : (B)]

DIBY-24

There are 200 individuals with skin disorder, out of them 100 have been exposed to Chemical C_1, 50 to Chemical C_2, and 30 to both C_1 & C_2. Find the number of individuals exposed to C_1 but not C_2.

[Ans. : 70]

DIBY-25

Evaluate the unknowns in each of the following cases:

(a) $n(A \cup B), n(A \cap B)$, **if** $A \subset B, n(A) = 30, n(B) = 36$

(b) $n(B)$, **if** $n(A) = 30, n(A \cup B) = 50, n(A \cap B) = 10$

(c) $n(A \cup B)$, **if** $n(A) = 20, n(A' \cap B) = 8$

(d) $n(A \cap B)$, **if** $n(A - B) = 30, n(B - A) = 16, n(A \cup B) = 60$

(e) $n(A \cup B)$, **if** $n(A) = 58, n(B) = 102, n(A - B) = 38$

(f) $n(A \cup B)$, **if** $n(S) = 70, n(A) = 40, n(B) = 50, n(A \cap \bar{B}) = 27$

(Here S denotes the Universal Set)

[Ans. (a) $n(A \cup B) = 36; n(A \cap B) = 30$
(b) $n(B) = 30$
(c) $n(A \cup B) = 28$
(d) $n(A \cap B) = 14$
(e) $n(A \cup B) = 140$
(f) data is incorrect.]

DIBY-26

A survey shows that $61\%, 46\%$ **and** 29% **of the people watched "3 idiots", "Rajneeti" and "Avatar" respectively.** 25% **people watched exactly two of the three movies and** 3% **watched none. What percentage of people watched all the three movies?**

[Ans. : 7]

DIBY-27

The set $(A \cap B')' \cup (B \cap C)$ is equal to

(A) $A' \cup B \cup C$ **(B)** $A' \cup B$ **(C)** $A' \cap C'$ **(D)** $A' \cap B$

[Ans. : (B)]

DIBY-28

Let X be the universal set for sets A and B, if $n(A) = 200$, $n(B) = 300$ and $n(A \cap B) = 100$, then $n(A' \cap B')$ is equal to 300 provided $n(X)$ is equal to

(A) 600 **(B)** 700 **(C)** 800 **(D)** 900

[Ans. : (B)]

DIBY-29

In a group of 50 people, two tests were conducted, one for diabetes and one for blood pressure. 30 people were diagnosed with diabetes and 40 people were diagonsed with high blood pressure. What is the minimum number of people who were having diabetes and high blood pressure?

(A) 0 **(B)** 10 **(C)** 20 **(d)** 30

[Ans. : (C)]

DIBY-30

A class has 175 students. The following data shows the number of students obtaining one or more subjects. Mathematics 100, Physics 70, Chemistry 40; Mathematics and Physics 30, Mathematics and Chemistry 28, Physics and Chemistry 23; Mathematics, Physics and Chemistry 18. How many students have offered Mathematics alone?

(A) 35 (B) 48 (C) 60 (D) 22

[Ans. : (C)]

DIBY-31

In a class of 80 students numbered 1 to 80, all odd numbered students opt of Cricket, students whose numbers are divisible by 5 opt for Football and those whose numbers are divisible by 7 opt for Hockey. If the number of students who do not opt any of the three games is n, then $\frac{n}{4}$ is equal to

[Ans. : 7]

It is never about end result

It's always about Journey

THANK You

#futureIITians